04. THE CIRCULATORY SYSTEM UPDATE

DURATION : 03:59 STUDY SHEET

COURSE SUMMARY

This video offers a profound update to the traditional model of the circulatory system, demonstrating that there are in fact three distinct venous systems. You will learn about the complex interactions between blood and the heart, as well as the different pathways blood can take, illustrating the intrinsic intelligence of circulation. The theories of pioneers such as M. Herveillat and M. Batson allow us to rethink the roles of veins and understand the cardinal, vitelline, and umbilical systems in detail. Practical implications for common pathologies, such as lower back pain and local acidity, will also be addressed, providing you with concrete tools to enrich your osteopathic practice.

THE CIRCULATORY SYSTEM: AN UPDATE

THE DEVELOPMENT OF VASCULAR PATHWAYS

The traditional image of the **circulatory system** is that of a circuit starting from the heart via the **arteries** and returning to it via the **veins**. While this is true, there are actually three different venous systems:

- **A vertebral system:** exclusively in relation to the vertebral system.
- **A portal system:** draining the entire **ectodermal**, **mesodermal**, and **endodermal** systems.
- **A third type of venous system.**

Blood does indeed return to the heart, but it can also turn back or take other pathways depending on the pressures and depressions at the heart. For example, if the portal system is overloaded and cannot ensure its return, the blood will take another route. It will use a **caval return system** via diaphragmatic exchange areas.

This means that pressures can adjust. Blood does not follow a simple closed circuit but can take various routes. There is an **intrinsic intelligence** to this system, always aiming to return to the heart.

A RENEWED VISION OF THE VENOUS SYSTEM

Blood does indeed leave the heart and return to it. However, our understanding of the venous system, and by extension the circulatory system, dates back to the 17th century. This conception has now been **renewed**.

The pioneers of this new understanding are **M. Herveillat** and **M. Batson**. A question arises: "But I was always told that veins contain valves to prevent blood from flowing backwards." In reality, these valves are only present in the **large veins**, and not throughout the entire network. There are very few small valves of this type.

THE DIFFERENT VENOUS SYSTEMS TO STUDY

We will study:

- **The cardinal veins:** which will become subcardinal and supracardinal.
- **The vitelline system:** in relation to the liver.
- **The umbilical system:** which will be remapped into the vertebral-caval-portal system and reintegrated into osteopathic practice.

The cardinal system will change so much that we will see in detail how it forms tissues and structures that will integrate into both **ectodermal** and **mesodermal** structures.

A method has been developed to put this into practice. It is crucial to have understood its meaning.

PRACTICAL AND CLINICAL IMPLICATIONS

Stasis and changes in **abdominal pressure** can have significant consequences. For example, **low back pain** can be the result of an overload of the portal system.

It is then necessary to "unload" the vertebral system due to this portal overload. This is where the entire mechanism of **discopathies** comes into play.

Any form of stasis will lead to **acidity**, even local, which will need to be resolved. The same pattern is always used. For example, a **sacral tumour** is almost always accompanied by **metastases** to the left kidney, left lung, and brain. A constant trajectory is observed.

In **prostate problems**, the same pattern can be found. Furthermore, there is **vascular continuity** between the vertebral sinuses and the sacral sinuses of the lesser pelvis.

A blood test taken from one arm does not reflect the totality of the blood that could be taken from a venous, arterial, or other system.



FIG. — Cardiovascular system

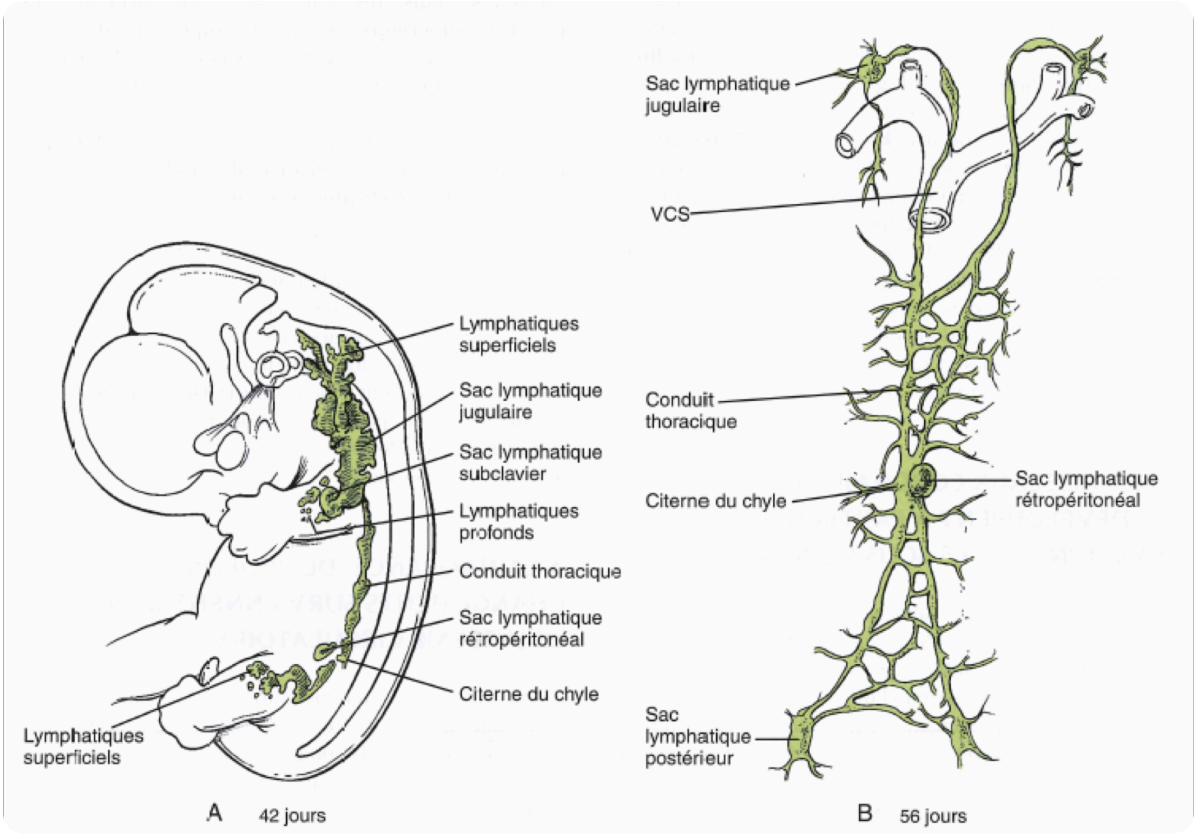


FIG. — Lymphatic system development

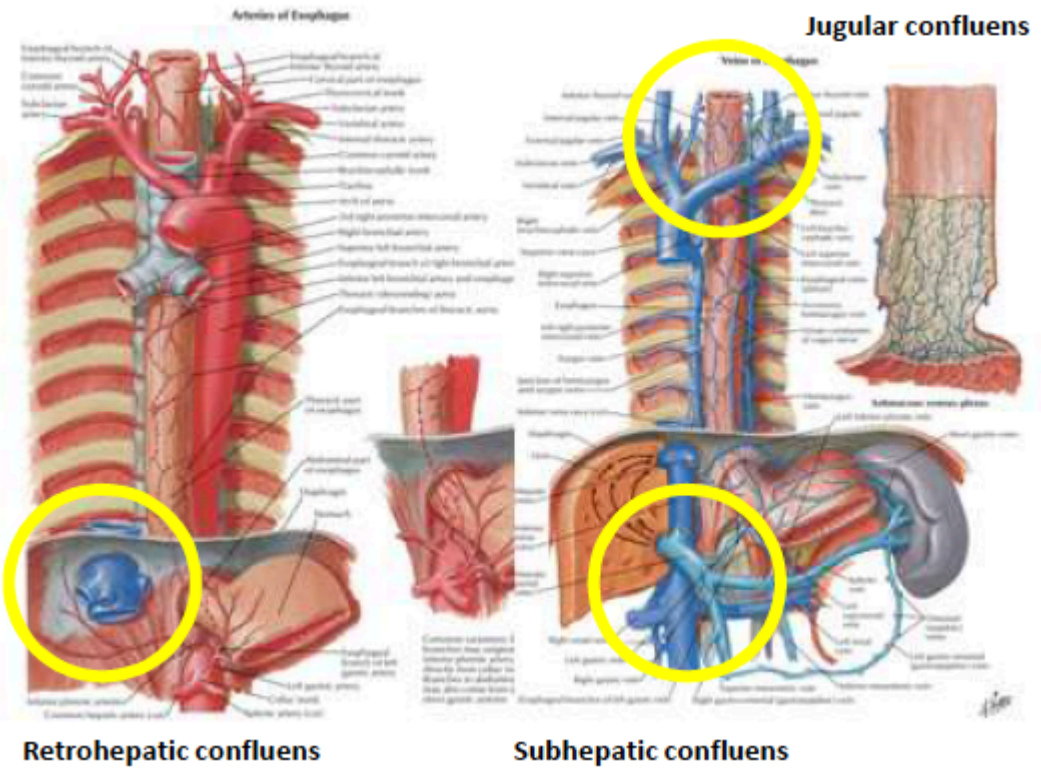


FIG. — Esophageal collateral veins

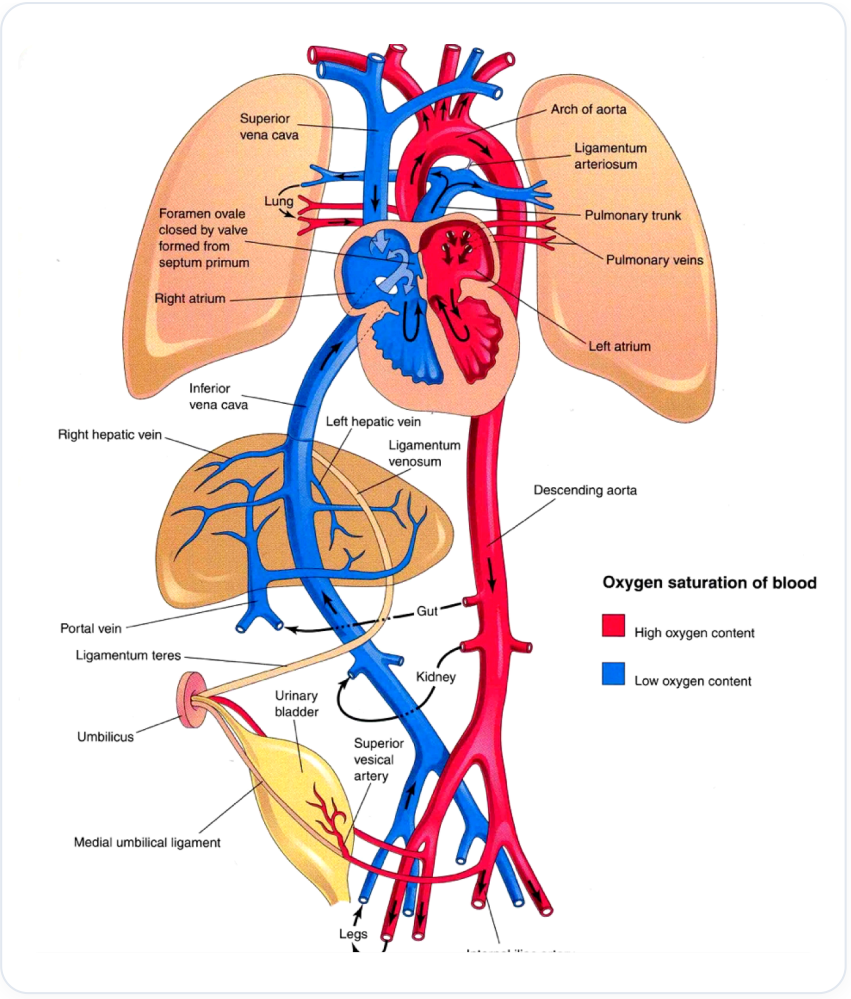


FIG. — Human fetal circulation